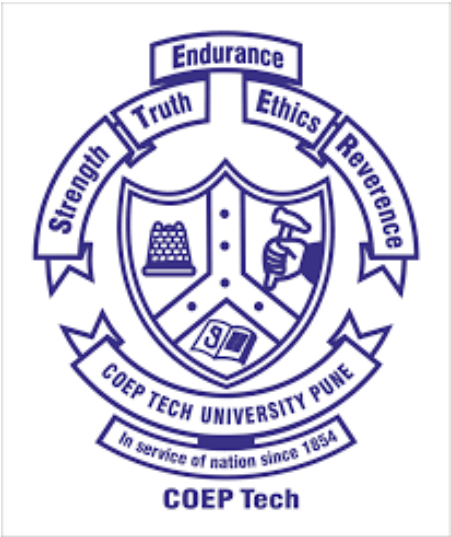


Project Stage 1 Report

Biopotential Signal Acquisition for Biomedical Application

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Abstract

This report outlines the preliminary signal acquisition of biopotential signals and the digital processing methodology. This is the first step in a series of planned project milestones, to ultimately achieve the task of controlling an "end application" using EEG signals. This project uses the BioAmp EXG Pill, an analog 2 channel biopotential signal amplifier and ESP32 microcontroller for signal acquisition. This is used to record EEG and EMG signals and determine the signal quality and processing techniques.

1 Introduction

Brain-Computer Interfaces (BCIs) provide a direct communication pathway between the human brain and external devices, bypassing traditional neuromuscular output channels. While traditional clinical EEG requires extensive electrode arrays and conductive gels, practical consumer-grade BCI applications demand systems that are unobtrusive, cost-effective, and easy to deploy.

2 Objectives

- Democratize BCI technology by validating a minimal-channel approach for control applications.
- Explore the EMG signal from the perspective of data acquisition and characterization.
- Explore the utility of the Prefrontal Cortex (positions Fp1 and Fp2) as a signal source.
- Create an accessible system: The forehead is hairless, allowing for rapid electrode placement and lower impedance contact compared to parietal or occipital regions.
- Signal Diversity: The PFC provides access to both "active" cognitive signals (Beta band activity during concentration) and "intentional" artifacts (Electrooculography or EOG from eye

3 Literature Review

3.1 Analog Front-End (AFE): BioAmp EXG Pill

The core of the acquisition system is the **BioAmp EXG Pill**, a compact biopotential amplifier based on the Texas Instruments **TL074H** Quad Low-Noise JFET-Input Operational Amplifier. This stage is responsible for differential amplification, noise rejection, and analog filtering.

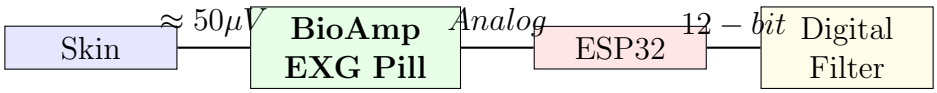


Figure 1: Signal Acquisition Chain Block Diagram.

3.1.1 Instrumentation Architecture

The module utilizes a customized instrumentation amplifier topology. Unlike standard single-ended amplifiers, this configuration amplifies the *difference* between the differential electrodes while rejecting common-mode noise (such as 50Hz mains hum) present on both inputs.

3.1.2 Key Technical Specifications

The specific operating parameters utilized in this project are detailed below. Note that the Input Impedance (Z_{in}) is critically high to prevent signal loading of the high-impedance skin-electrode interface.

Parameter	Value	Significance in EEG
Signal Gain	1000 V/V	Amplifies 50 μV brain signals to $\approx 50mV$ range for detection.
Input Impedance	$10^{12}\Omega$ (1 TΩ)	Prevents signal attenuation due to skin resistance.
CMRR	$>$ 100 dB	Rejects electrical interference common to both inputs.
Bandwidth	0.5 – 40 Hz	Hardware-limited to reject DC drift and high-frequency muscle noise.
Noise Floor	< 1 μV RMS	Ensures the noise is smaller than the target Alpha/Beta waves.
Op-Amp Type	JFET Input	Reduces input bias current, minimizing artifacts.

Table 1: BioAmp EXG Pill Hardware Specifications.

3.1.3 Driven Right Leg (DRL)

The BioAmp module incorporates a "Driven Right Leg" circuit (via the REF pin connected to the Mastoid). This active feedback loop inverts the common-mode noise measured at the inputs and injects it back into the body, effectively cancelling out interference before amplification.

Following analog conditioning, the signal is digitized by the ESP32 microcontroller. The specific configuration used ensures Nyquist criterion compliance and optimal resolution.

3.1.4 Analog-to-Digital Converter (ADC)

- **Resolution:** 12-bit (Values 0 – 4095).
- **Input Voltage Range:** 0V – 3.3V.
- **LSB Sensitivity:** The voltage represented by a single digital step is calculated as:

$$V_{LSB} = \frac{V_{ref}}{2^N} = \frac{3.3V}{4096} \approx 0.805mV \tag{1}$$

  Given the AFE gain of 1000x, this provides an effective EEG resolution of approximately $0.8\mu V$ at the scalp, sufficient for detecting Beta band activity.

3.1.5 Sampling Rate

The firmware is configured for a fixed sampling rate (F_s) of **250 Hz**.

- **Nyquist Limit:** $F_{nyquist} = F_s/2 = 125Hz$.
- **Justification:** Since the EEG spectral content of interest (Beta waves) extends only to 30 Hz, a 250 Hz sampling rate provides ample oversampling (approx 8x), allowing for superior digital filtering performance and minimal aliasing.

3.2 Introduction to Prefrontal EEG

Electroencephalography (EEG) measures the aggregate electrical activity of the cerebral cortex via electrodes placed on the scalp. The signal represents the summation of excitatory and inhibitory postsynaptic potentials generated by pyramidal neurons.

In this study, recording is focused on the **Prefrontal Cortex (PFC)**, specifically utilizing the **Fp1** (Left Frontopolar) and **Fp2** (Right Frontopolar) positions. These locations are situated directly above the orbits (eye sockets) and cover the anterior frontal lobe.

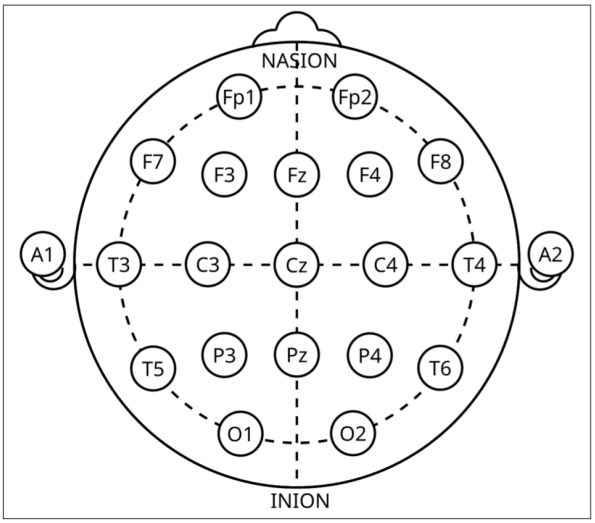


Figure 2: The International 10-20 System standardizing electrode placement. Fp1 and Fp2 represent the Frontopolar sites.

3.3 Physiology of the Prefrontal Cortex

The Prefrontal Cortex is phylogenetically one of the most recently developed regions of the mammalian brain. Its activity is distinct from sensory or motor cortices.

3.3.1 Functional Neuroanatomy

The underlying neural tissue at Fp1 and Fp2 is responsible for "Executive Functions." Key physiological roles include:

- **Cognitive Control:** Regulation of thought processes, attention allocation, and mental flexibility.
- **Working Memory:** The ability to temporarily hold and manipulate information (e.g., during mental arithmetic or logical reasoning).

3.3.2 Signal Artifact Susceptibility

Due to the anatomical proximity of Fp1 and Fp2 to the eyes and facial muscles, signals recorded here are heavily influenced by non-cerebral potentials:

1. **Ocular Artifacts (EOG):** The eye acts as a dipole (positive cornea, negative retina). Vertical eye movements (blinks) and horizontal saccades generate high-amplitude potentials (Electrooculogram) that overlay the EEG signal.

2. **Muscle Artifacts (EMG):** Contraction of the *frontalis* (forehead) and *temporalis* (jaw) muscles introduces high-frequency noise.

3.3.3 Frequency Domain Analysis

EEG signals are non-stationary and are typically analyzed by decomposing them into specific frequency bands. The spectral power distribution at the Prefrontal Cortex varies significantly based on cognitive state.

Band	**Frequency**	**Characteristics at Prefrontal Cortex (Fp1/Fp2)**
Delta	0.5 – 4 Hz	**Physiological:** Dominant during deep (NREM) sleep in healthy adults. **Artifactual:** In waking states at Fp1/Fp2, this band is dominated by ocular artifacts (blinks and eye movements), often exceeding 100 μV.
Theta	4 – 8 Hz	**Physiological:** Associated with drowsiness, day-dreaming, or deep meditation. **Cognitive:** Frontal midline theta can increase during high working memory load (e.g., n-back tasks).
Alpha	8 – 13 Hz	**Physiological:** Prominent during wakeful relaxation with eyes closed. **Topography:** Typically strongest in the occipital lobe; however, "Alpha Propagation" allows it to be detected frontally. It attenuates (Alpha Blocking) upon opening eyes or focusing attention.
Beta	13 – 30 Hz	**Physiological:** The dominant rhythm of active waking consciousness. **Cognitive:** Power increases bilaterally at Fp1/Fp2 during intense concentration, analytical problem solving, and anxiety.
Gamma	$>$ 30 Hz	**Physiological:** Associated with cross-modal sensory processing and short-term memory matching. **Artifactual:** This band is frequently contaminated by electromyographic (EMG) noise from facial tension.

Table 2: Spectral characteristics of EEG signals recorded at Frontopolar sites.

4 Methodology

- **Data Acquisition:** sEMG signals were recorded at a sampling rate of ≈ 250 Hz under three conditions:
    1. **Fast:** Rapid, unloaded movement.
    2. **Slow:** Controlled, unloaded movement.
    3. **Weight:** Loaded movement to induce high motor unit recruitment.
- **E**EG signals were recorded at the FP1 and FP2 locations at a sampling rate of ≈ 250 Hz with the following actions:
    1. Idle state
    2. Eyes closed
    3. Blink thrice
    4. Lift right arm
    5. Think about lifting right arm
    6. Move eyes to the left and right each and centre for 30 seconds and centre them.
- **Preprocessing Pipeline (MATLAB):**
    - **DC Offset Removal:** Zero-centering the signal.
    - **Notch Filter (48-52 Hz):** 2nd-order Butterworth to remove power line interference.
    - **Bandpass Filter (20-180 Hz):** 4th-order Butterworth to isolate motor unit action potentials.
    - **Envelope Detection:** Full-wave rectification + 5Hz Low-pass filter.

5 Acquiring and Analyzing EMG Signals

In this section, Surface Electromyography (sEMG) data under varying conditions (Fast, Slow, Weighted) are observed. A signal processing pipeline is designed for artifact removal and the frequency domain characteristics of muscle movement. This experiment serves as a validation step for a high-sensitivity EEG acquisition system.

5.1 Signal Verification & Filtering

Observation: The raw signal exhibited significant DC drift and high-frequency noise. The filtering pipeline successfully stabilized the baseline (centered at 0 mV) and removed artifacts while preserving the muscle burst morphology.

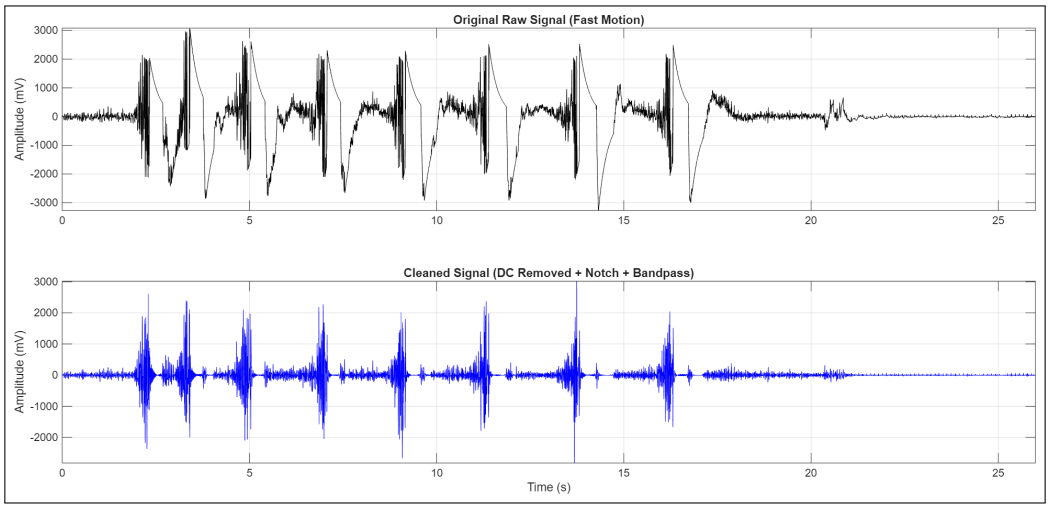


Figure 3: Comparison of Raw Signal (Top) vs. Processed Signal (Bottom).

5.2 Time Domain Analysis (Envelope)

Observation: The envelope analysis reveals distinct activation patterns. The **Weighted** condition (Red) shows the highest amplitude peaks, indicating maximum motor unit recruitment. The **Slow** condition (Blue) demonstrates lower amplitude but wider activation durations compared to **Fast** (Green).

5.3 Frequency Domain Analysis (PSD)

Observation: The Power Spectral Density (PSD) plot shows the energy distribution of the signal. The **Weighted** signal (Red) exhibits significantly higher power (dB) across the 20-100Hz band compared to the unloaded **Fast** signal (Green). A distinct drop at 50Hz confirms the successful operation of the Notch filter.

5.4 Time-Frequency Analysis (Spectrogram)

Observation: The spectrogram visualizes the signal intensity over time. Vertical bands of high activity (yellow/orange) correspond to the concentric contraction phases of the

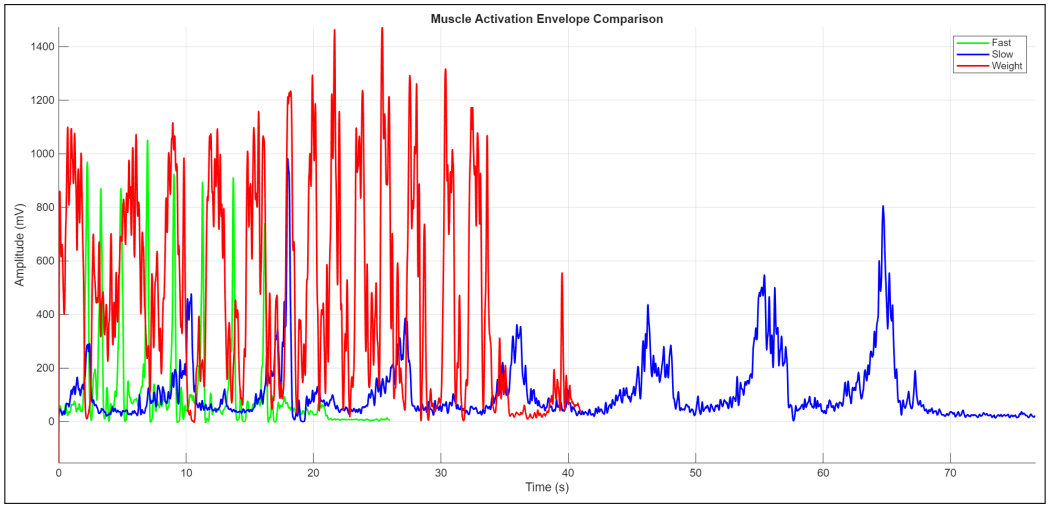


Figure 4: Envelope comparison showing amplitude differences across conditions.

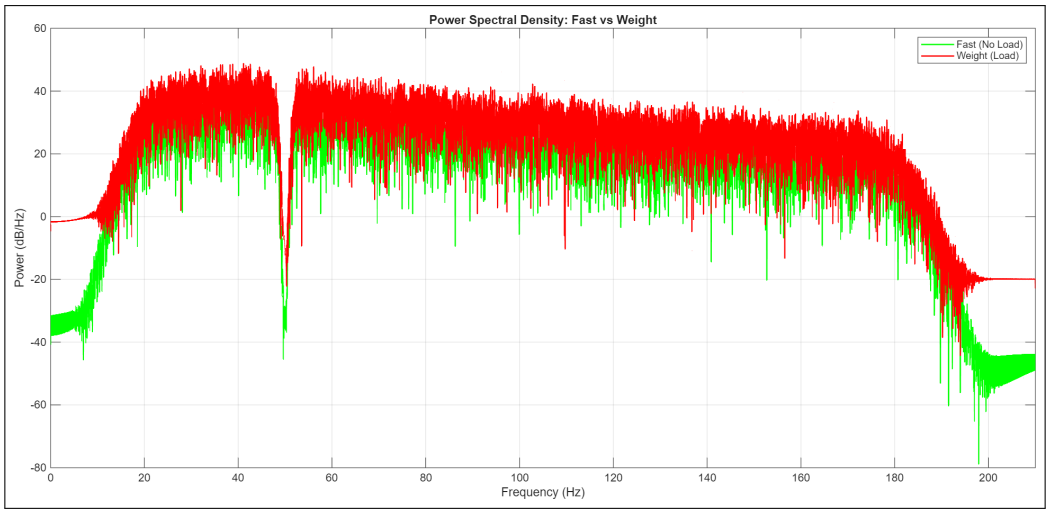


Figure 5: Power Spectral Density comparison. Note the higher energy output in the loaded state.

exercise. The frequency content is concentrated between 20Hz and 100Hz, consistent with standard sEMG characteristics.

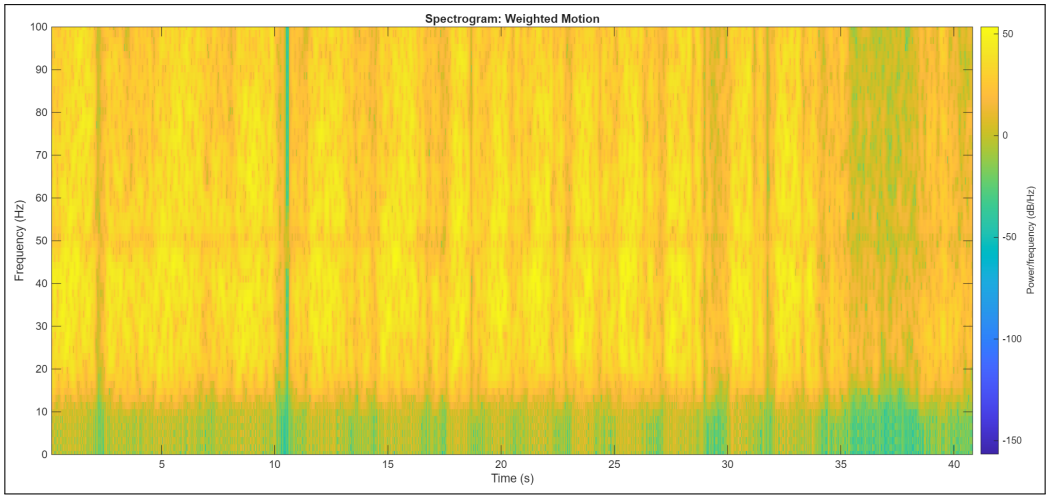


Figure 6: Spectrogram of Weighted Data showing transient muscle events.

5.5 Quantitative Metrics (RMS)

Observation: Root Mean Square (RMS) values confirm the visual analysis. The **Weight** condition resulted in an RMS roughly double that of the unloaded conditions, providing a quantifiable metric for "Muscle Effort."

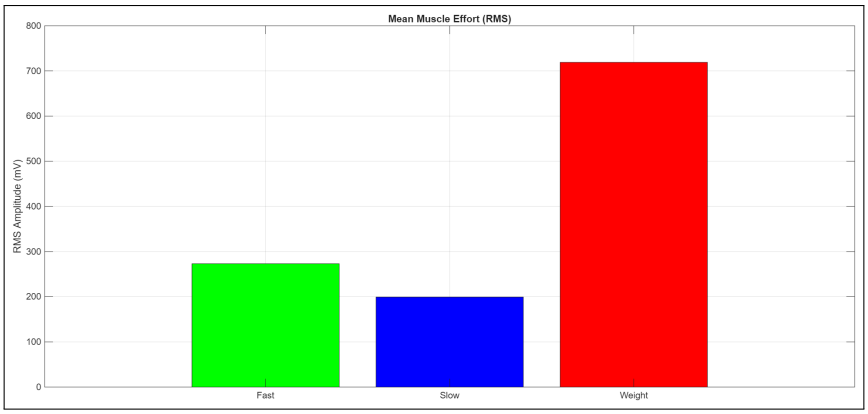


Figure 7: Mean RMS values comparing muscle effort across the three datasets.

6 Acquiring and Analysing Prefrontal EEG Signals

6.1 Baseline Calibration (Idle State)

Time Domain: The "Idle" dataset establishes the system's noise floor at approx $\pm 50\mu V$ (Figure 8). **Frequency Domain:** The Power Spectral Density (PSD) in Figure 9 shows a typical $1/f$ noise distribution with a distinct mains hum spike at 50Hz, validating the need for the digital Notch filter implemented in the MATLAB pipeline.

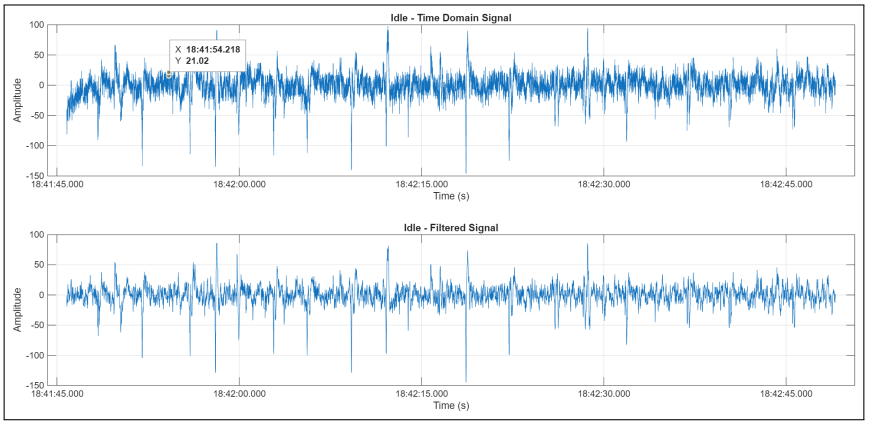


Figure 8: **Idle State (Time):** Establishing the noise floor.

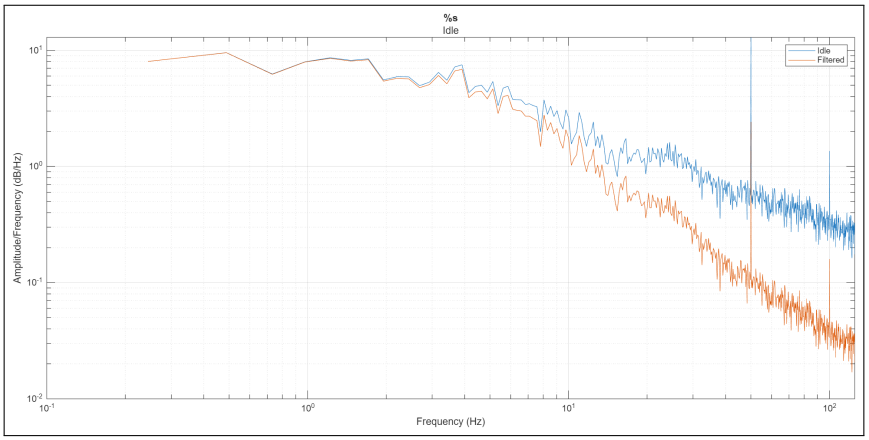


Figure 9: **Idle State (PSD):** Frequency baseline check.

6.2 Alpha Wave Detection (Eyes Closed)

Observation: To detect Alpha waves (8-12 Hz), data was recorded with eyes closed. Figure 10 shows the time domain signal, while Figure 11 shows the spectral power. While Fp1/Fp2 is not the optimal location for Alpha waves (which are occipital), the PSD allows us to check for any low-frequency power increases compared to the Idle state.

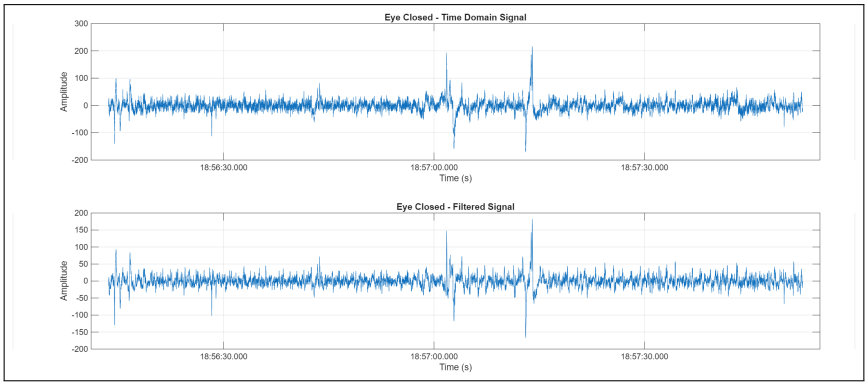


Figure 10: **Eyes Closed (Time):** Signal during relaxation.

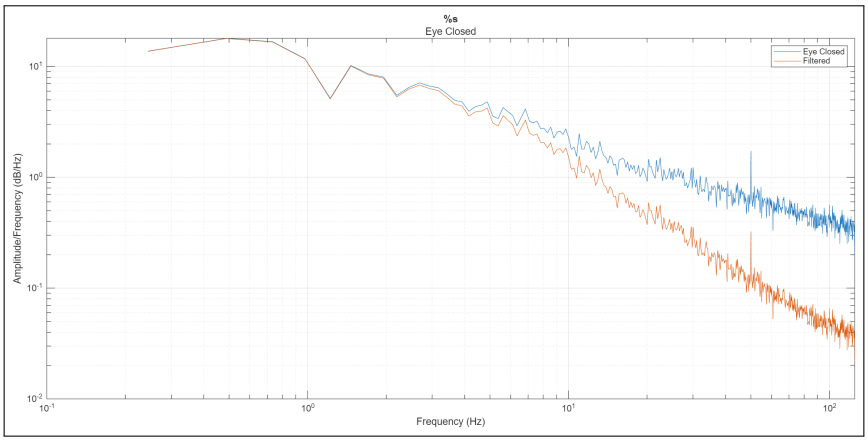


Figure 11: **Eyes Closed (PSD):** Spectral check for Alpha band activity.

6.3 Ocular Artifacts (Control Signals)

6.3.1 Triple Blink (Throttle)

Observation: The Triple Blink (Figure 12) generates massive spikes in the time domain. Figure 13 confirms that these artifacts are concentrated in the **Delta Band (0.5 - 4 Hz)**. The high energy density in low frequencies differentiates real blinks from high-frequency muscle noise.

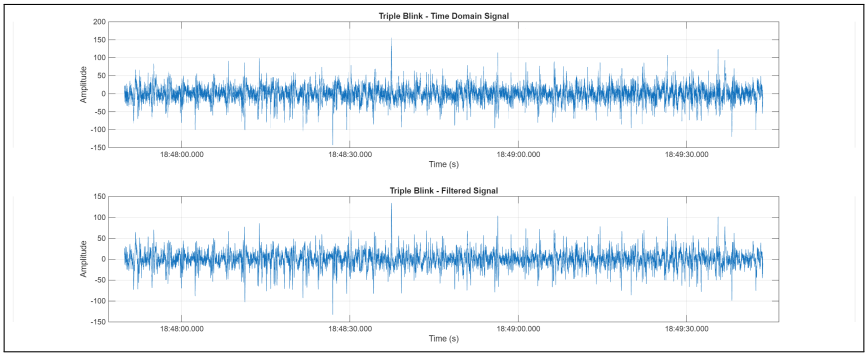


Figure 12: **Triple Blink (Time):** High amplitude spikes for binary triggering.

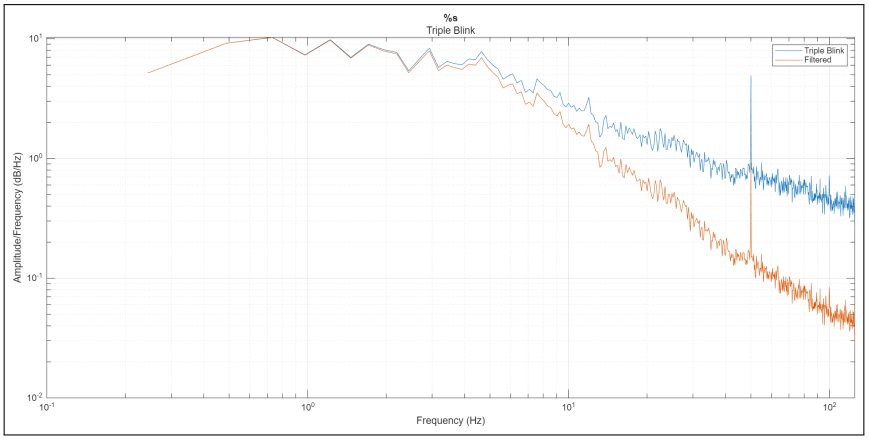


Figure 13: **Triple Blink (PSD):** High energy density in low frequencies.

6.3.2 Vertical/Lateral Gaze

Observation: Figure 14 depicts the "30L/30R" (Lateral Gaze) sequence. The PSD analysis in Figure 15 confirms that the DC polarity shifts seen in the time domain correspond to ultra-low frequency components (< 1 Hz), further validating the EOG nature of the signal.

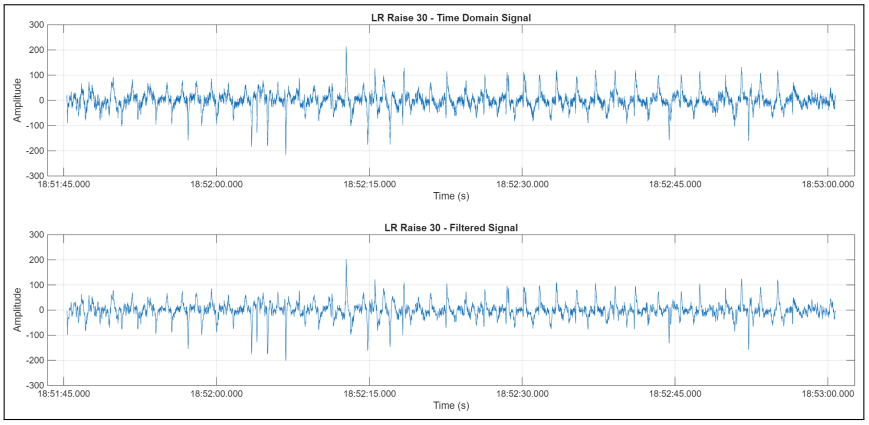


Figure 14: **Gaze Calibration (Time):** Signals showing polarity drift.

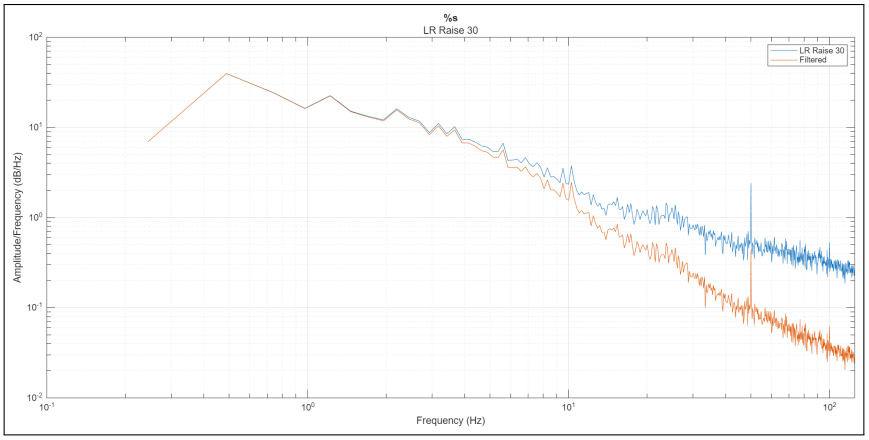


Figure 15: **Gaze Calibration (PSD):** Dominance of low-frequency EOG components.

6.4 Motor Imagery Validation (Frequency Domain)

Observation: To verify if motor commands could be detected at the forehead, we compared "Thinking" about movement versus "Performing" movement. Figure 17 (Raise Right Hand) shows significantly higher high-frequency noise (> 20 Hz) compared to Figure 16 (Think Right Hand).

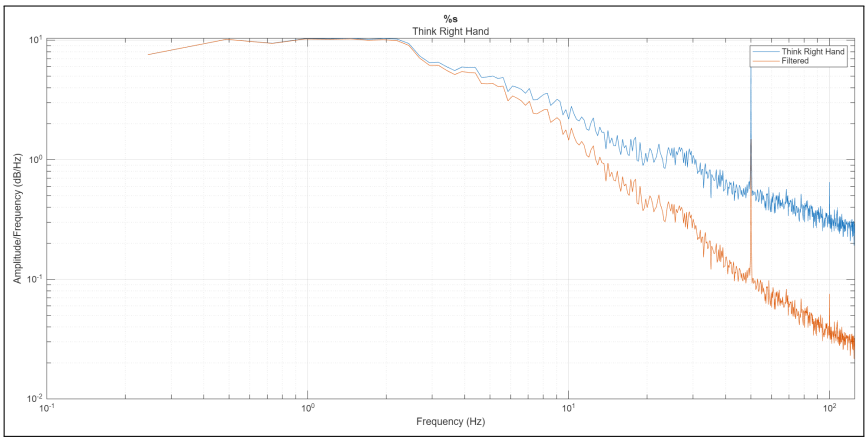


Figure 16: **Think Right Hand (PSD):** Lack of significant motor features.

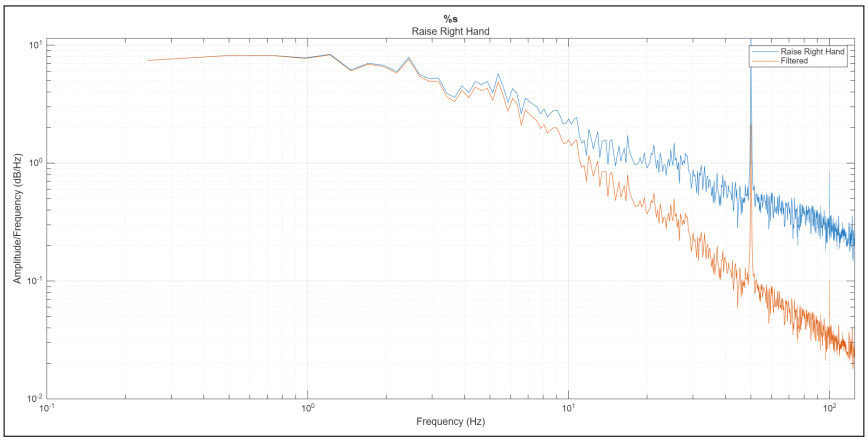


Figure 17: **Raise Right Hand (PSD):** Presence of high-frequency EMG artifacts.

6.5 Extended Signal Characterization

To complete the dataset analysis, the remaining visualizations characterize the time-domain features of motor tasks, the spectral content of cognitive load, and the system verification steps.

Cognitive Analysis (Calc): Figure 18 and Figure 19 represent the "Mental Math" task. Unlike the EOG signals, the time domain is noisy and complex. The PSD is required to isolate the Beta band (13-30Hz) activity indicative of concentration.

Motor Time Domain: Figure 20 (Thinking) shows a stable baseline similar to the Idle state, confirming that imagined movement does not generate artifacts. In contrast, Figure 21 (Raising) shows jagged spikes, proving that physical movement introduces EMG noise from facial muscles.

System Verification: Finally, the data integrity was confirmed via the dataset structure (Figure **??**), real-time visualization in Spike Recorder (Figure **??**), and a bulk processing verification plot (Figure **??**) to ensure all CSVs were readable.

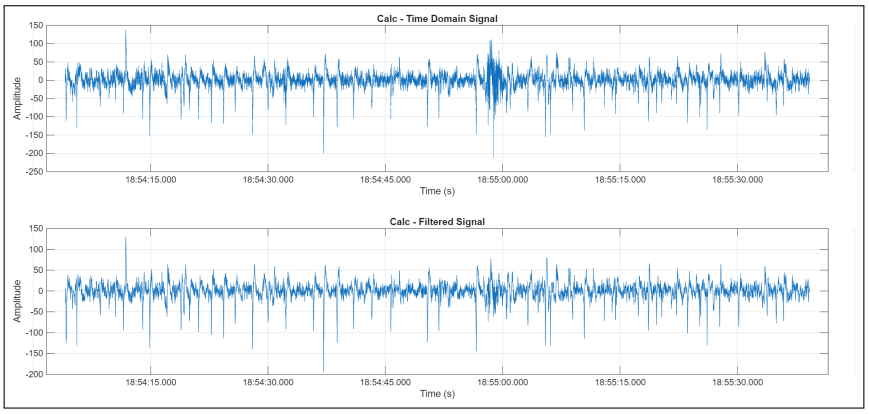


Figure 18: **Mental Math (Time):** Complex waveform during calculation.

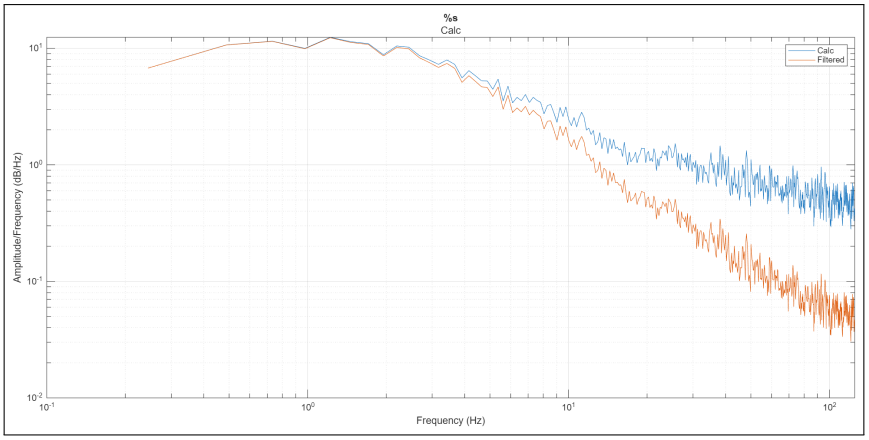


Figure 19: **Mental Math (PSD):** Spectral density showing Beta activity.

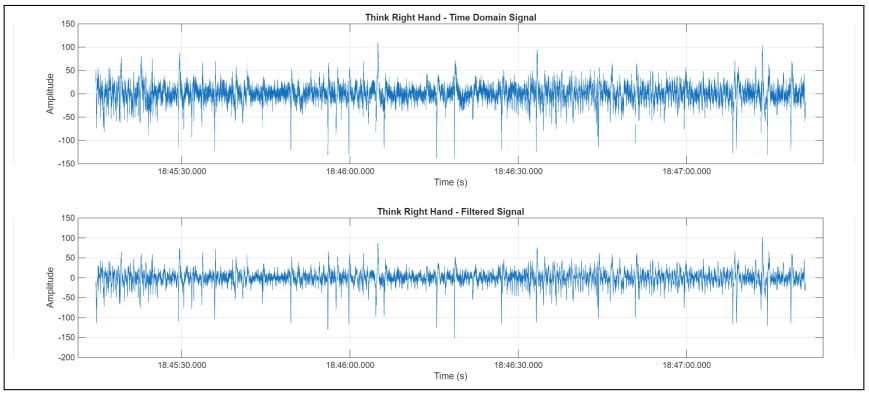


Figure 20: **Think Right Hand (Time):** Stable baseline during imagery.

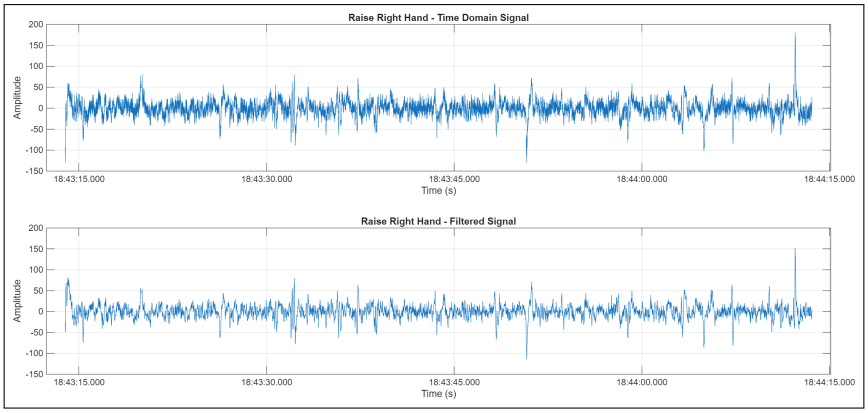


Figure 21: **Raise Right Hand (Time):** EMG artifacts from physical movement.

7 Results and Conclusion

The objective of Project Stage I was to validate the signal acquisition chain and characterize the bio-potentials recordable from the Prefrontal Cortex (Fp1/Fp2). The experimental analysis yields the following conclusions:

- **Signal Integrity:** The BioAmp EXG system successfully captures surface potentials with a noise floor of approximately $\pm 50\mu V$. The implementation of digital filtering (Notch and Bandpass) effectively isolates the physiological bands of interest from environmental interference.
- **Control Signal Viability:** The most robust signals for vehicle control were identified as Electrooculography (EOG) artifacts.
  - **Blinks:** Generate high-amplitude ($> 200\mu V$), low-frequency spikes ideal for binary switching (Throttle).
  - **Lateral Gaze:** Produces distinct DC polarity shifts suitable for directional control (Steering).
- **Cognitive State Detection:** While Beta-band activity (13-30 Hz) was successfully detected via PSD analysis during mental arithmetic, the signal amplitude is significantly lower than EOG artifacts, requiring precise spectral thresholding for reliable detection.
- **Motor Imagery Constraints:** Time-domain analysis confirmed that motor imagery (thinking about hand movement) does not generate distinguishable signals at the Fp1/Fp2 location. Any "movement" signals detected were identified as EMG artifacts from facial muscle tension.

In conclusion, the acquisition system is verified as functional. The control logic for the next stage will prioritize EOG artifacts for "Hard" control (Steering/Motion) due to their high Signal-to-Noise Ratio (SNR), while exploring Beta-band power for "Soft" control (Speed modulation).

8 Limitations

Despite the successful acquisition of valid EEG/EOG data, the current Stage I prototype presents specific constraints:

1. **Spatial Resolution:** The current setup utilizes a single differential channel (Fp1 minus Fp2). This "lumps" the activity of the left and right prefrontal hemispheres into a single voltage difference, making it impossible to distinguish lateralized brain activity or localize signal sources with high precision.

2. **Artifact Susceptibility:** Being located on the forehead, the electrodes are positioned directly over the *frontalis* muscles. Unintended facial expressions (frowning, raising eyebrows) generate Electromyographic (EMG) noise that overlaps with the EEG frequency spectrum, potentially causing false triggers in the control logic.

3. **Mobility Constraints:** The current data logging relies on a wired USB serial connection to a PC. This "tethering" restricts the user's range of motion, which is a significant bottleneck for controlling a remote vehicle in a real-world scenario.

9 Future Scope

To address the identified limitations and transition from signal acquisition to a practical BCI vehicle controller, the following developments are proposed for Stage II:

9.1 Hardware Design for Specific Application

The current breadboard/module-based prototype will be miniaturized into a custom Printed Circuit Board (PCB). This design will integrate the Analog Front End (AFE) directly with the ESP32 and a battery management system to create a standalone, wearable headset form factor, eliminating the need for wired PC tethering.

9.2 Increasing Spatial Resolution

To enable more intuitive control schemes (such as true Motor Imagery), the system will be expanded to include additional electrode channels. Specifically, targeting the Motor Cortex positions (**C3 and C4**) would allow for the detection of Mu rhythms (8-13 Hz desynchronization), enabling the user to control the vehicle by imagining hand movements rather than relying solely on eye movements.

9.3 Long Duration Data Machine Learning

Future work will involve collecting longer-duration datasets (¿ 30 minutes) to analyze signal drift and user fatigue. This data will be used to train lightweight Machine Learning classifiers on the microcontroller. These algorithms will aim to distinguish between "Intentional Control Commands" and "Unintentional Fatigue Artifacts," thereby improving the robustness and safety of the vehicle control logic.